MA10509: Mathematics for Data Science

Unit 5: Data Science Tools and Case Studies

List of R-Programs

1. Display “Hello World”

```
# Program to print a message
print("Hello, World!")
```

2. Add two numbers

```
# Program to add two numbers
a <- 10
b <- 5
sum <- a + b
print(paste("Sum =", sum))
```

3. Find square and cube of a number

```
num <- 4
square <- num^2
cube <- num^3
print(paste("Square:", square))
print(paste("Cube:", cube))
```

4. Find square and cube of a number

```
num <- 4
square <- num^2
cube <- num^3
print(paste("Square:", square))
print(paste("Cube:", cube))
```

5. Find largest of three numbers

```
a <- 12
b <- 25
c <- 9
if(a > b & a > c){
  print("a is largest")
} else if(b > c){
  print("b is largest")
} else {
  print("c is largest")
}
```

6. Calculate area of circle

```
r <- 3
area <- pi * r^2
print(paste("Area of circle =", area))
```

7. Print numbers from 1 to 10

```
for(i in 1:10){
print(i)
}
```

8. Sum of first N natural numbers

```
n <- 10
sum <- 0
for(i in 1:n){
  sum <- sum + i
}
print(paste("Sum =", sum))
```

9. Generate multiplication table

```
num <- 5
for(i in 1:10){
  print(paste(num, "x", i, "=", num*i))
}
```

10. Find factorial of a number

```
n <- 5
fact <- 1
for(i in 1:n){
  fact <- fact * i
}
print(paste("Factorial =", fact))
```

11. Calculate Simple Interest

```
P <- 1000
R <- 5
T <- 2
SI <- (P * R * T) / 100
print(paste("Simple Interest =", SI))
```

12. Check if a number is Prime

```
n <- 7
flag <- TRUE
if(n > 1){
  for(i in 2:(n-1)){
    if(n %% i == 0){
      flag <- FALSE
      break
    }
  }
```

13. Create and display a vector

```
v <- c(2, 4, 6, 8, 10)
print(v)
```

14. Find Maximum and Minimum of a vector

```
nums <- c(3, 9, 5, 12, 7)
print(paste("Maximum =", max(nums)))
print(paste("Minimum =", min(nums)))
```

15. Display elements of a matrix

```
m <- matrix(1:9, nrow=3, ncol=3)
print("Matrix:")
print(m)
```

16. Calculate mean of numbers

```
nums <- c(10, 20, 30, 40, 50)
mean_value <- mean(nums)
print(paste("Mean =", mean_value))
```

17. Use a function to add two numbers

```
add <- function(a, b){
  return(a + b)
}
result <- add(5, 7)
print(paste("Sum =", result))
```

18. Plot a Simple Graph

```
x <- c(1,2,3,4,5)
y <- c(2,4,6,8,10)
plot(x, y, type="o", col="blue",
main="Simple Line Plot")
```

19. Plot a Bar graph

```
subjects <- c("Math", "Science",
"English")
marks <- c(85, 90, 80)
barplot(marks, names.arg=subjects,
col="skyblue", main="Marks in
Subjects")
```

20. Plot a Histogram

```
data <- c(2,3,3,4,4,4,5,5,6)
hist(data, main="Histogram Example",
col="lightgreen")
```